Engineering - Graduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Find the Laplace transform L{g(t)}, where g(t)= 0,t $<$ 4 and= (t - 4)^2,t $\geq$ 4

(2e^-4s) / s^3

   (a) 2 / s^3
   (b) (2 / s^2) * e^-4s
   (c) 4 / s^3

(2e^-4s) / s^4

2. A short dipole has a radiation resistance R_r=$\sqrt{}$(μ_0 / ϵ_0) [()^2 / 6π] ohms. FindA_emthe maximum effectiveaperture of this dipole.

   (a) 0.075 λ^2
   (b) 0.200 λ^2
   (c) 0.180 λ^2
   (d) 0.089 λ^2
   (e) 0.102 λ^2

3. y_11 = 0.25 mho, y_12 = - 0.05 mho, y_22 = 0.1 mho. Find the z-parameters.

   (a) z_11 = 11.1 Ω, z_12 = z_21 = 4.44 Ω, z_22 = 22.2 Ω
   (b) z_11 = 11.1 Ω, z_12 = z_21 = 2.22 Ω, z_22 = 4.44 Ω
   (c) z_11 = 2.22 Ω, z_12 = z_21 = 4.44 Ω, z_22 = 11.1 Ω
   (d) z_11 = 4.44 Ω, z_12 = z_21 = 1.11 Ω, z_22 = 22.2 Ω
   (e) z_11 = 4.44 Ω, z_12 = z_21 = 2.22 Ω, z_22 = 22.2 Ω

4. What percent of the total volume of an iceberg floats above the water surface ? Assume the density of ice to be 57.2lb_m/ft^3, the density of water 62.4lb_m/ft^3.

   (a) 5 percent
   (b) 40 percent
   (c) 35 percent
   (d) 10 percent
   (e) 20 percent

5. Let A =|22|,B =|31|. Find AB. |12||12| |42|

   (a) AB = |76| |75| |168|
   (b) AB = |96| |65| |158|

(c) AB = |76| |45| |138|

(d) AB = |96| |55| |138|

(e) AB = |86| |45| |138|

True / False (10 marks)

Answer True or False

6. True or False: The Laplace transform of f(t) = t^n is n! / s^n, which incorrectly omits a factor of s in the denominator.

7. True or False: A phase shift of 30° in a phase-shift keying system ensures synchronization, resulting in lower error probabilities than 10^-5.

8. True or False: The electrostatic energy of a point charge within a conducting shell is calculated using (q^2 / (4_0))(1/r_1 - 1/r_2).

9. True or False: The one cycle surge current rating of an SCR with a half cycle surge current of 3000 A at 50 Hz is 6000 A.

10. True or False: The curl of the vector field a = x^2y^2i + 2xyzj + z^2k at the point (1, -2, 1) is (0, 4, 0).

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the number of isotopes of an element influences the determination of the number of rings in the Bohr model, and analyze its implications for engineering applications.

12. Evaluate the surface integrals $\int\int_s F \cdot \hat{n}\, ds$ and $\int\int_s F \times \hat{n}\, ds$ for the vector field $F = kr^n\hat{r}$ over a sphere of radius a centered at the origin. Discuss the implications of your results.

End of Paper